
Computer Architecture

Memory System

Internal Memory

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- Dynamic RAM

Semiconductor Memory

- Semiconductor chips are used to store the data.
- The basic element is called as memory cell.
- They have two stable states(binary 1 and 0).

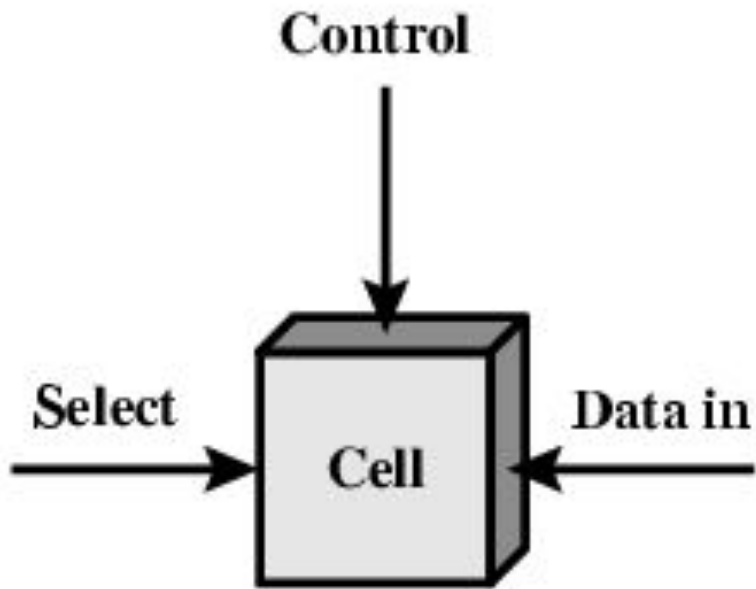
Semiconductor Memory Types

Memory Type	Category	Erasure	Write Mechanism	Volatility
Random-access memory (RAM)	Read-write memory	Electrically, byte-level	Electrically	Volatile
Read-only memory (ROM)	Read-only memory	Not possible	Masks	Nonvolatile
Programmable ROM (PROM)			Electrically	
Erasable PROM (EPROM)	Read-mostly memory	UV light, chip-level		
Electrically Erasable PROM (EEPROM)		Electrically, byte-level		
Flash memory		Electrically, block-level		

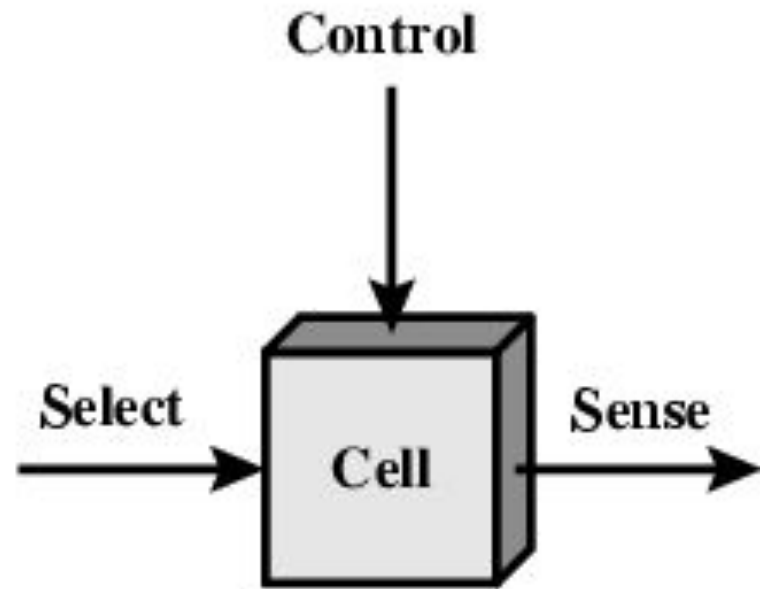
Semiconductor Memory

- RAM
 - Misnamed as all semiconductor memory is random access
 - Read/Write
 - Volatile
 - Temporary storage
 - Static or dynamic

Memory Cell Operation



(a) Write



(b) Read

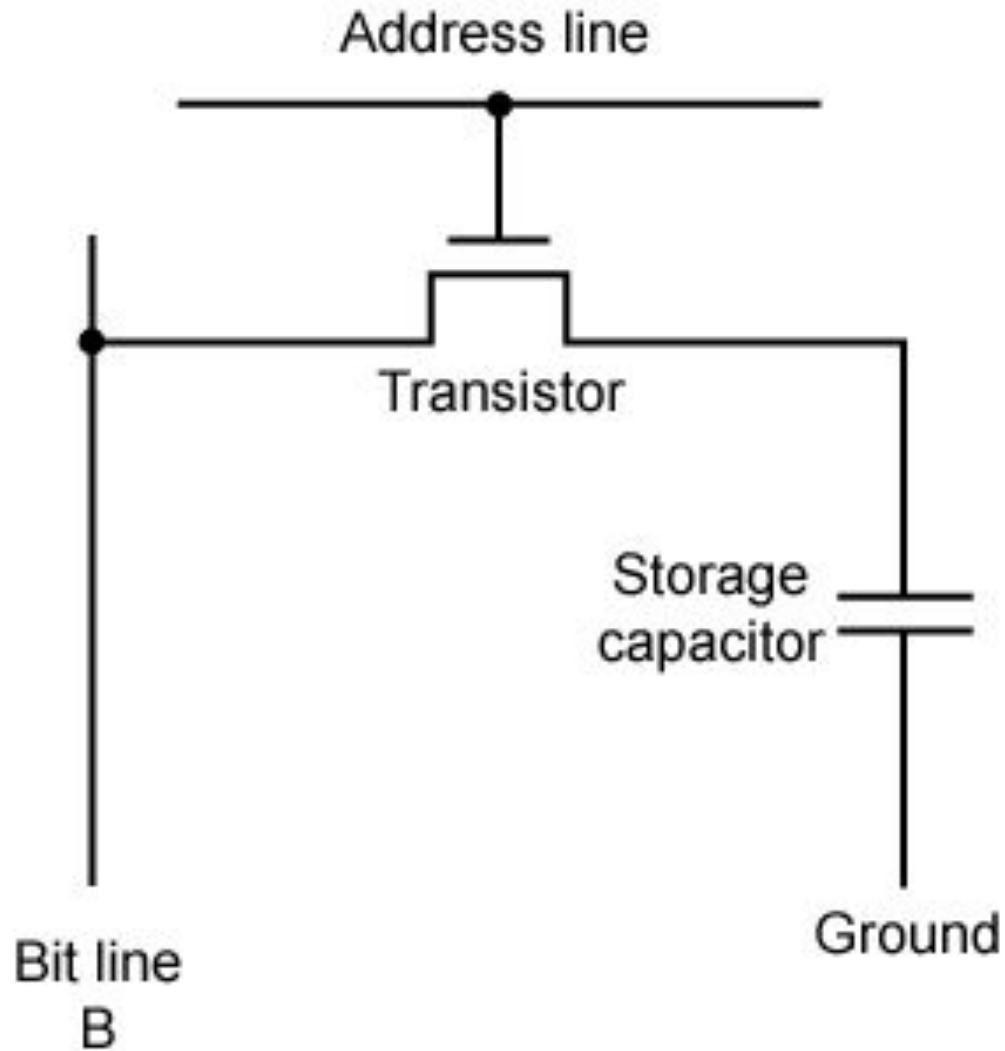
DRAM Operation

- Address line active when bit read or written
 - Transistor switch closed (current flows)
- Write
 - Voltage to bit line
 - High for 1 low for 0
 - Then signal address line
 - Transfers charge to capacitor
- Read
 - Address line selected
 - transistor turns on
 - Charge from capacitor fed via bit line to sense amplifier
 - Compares with reference value to determine 0 or 1
 - Capacitor charge must be restored

Dynamic RAM

- Bits stored as charge in capacitors
- Charges leak
- Need refreshing even when powered
- Simpler construction
- Smaller per bit
- Less expensive
- Need refresh circuits
- Slower
- Main memory
 - Level of charge determines value

Dynamic RAM Structure



Static RAM

- Bits stored as on/off switches
- No charges to leak
- No refreshing needed when powered
- More complex construction
- Larger per bit
- More expensive
- Does not need refresh circuits
- Faster
- Cache
- Digital
 - Uses flip-flops

SRAM v DRAM

- Both volatile
 - Power needed to preserve data
- Dynamic cell
 - Simpler to build, smaller
 - More dense
 - Less expensive
 - Needs refresh
 - Larger memory units
- Static
 - Faster
 - Cache

Read Only Memory (ROM)

- Permanent storage
 - Nonvolatile
- Microprogramming (see later)
- Systems programs (BIOS)
- Function tables

Types of ROM

- Written during manufacture
 - Very expensive for small runs
- Programmable (once)
 - PROM
 - Needs special equipment to program
- Read “mostly”
 - Erasable Programmable (EPROM)
 - Erased by UV
 - Electrically Erasable (EEPROM)
 - Takes much longer to write than read
 - Flash memory
 - Erase whole memory electrically

Refreshing

- Refresh circuit included on chip
- Disable chip
- Count through rows
- Read & Write back
- Takes time
- Slows down apparent performance

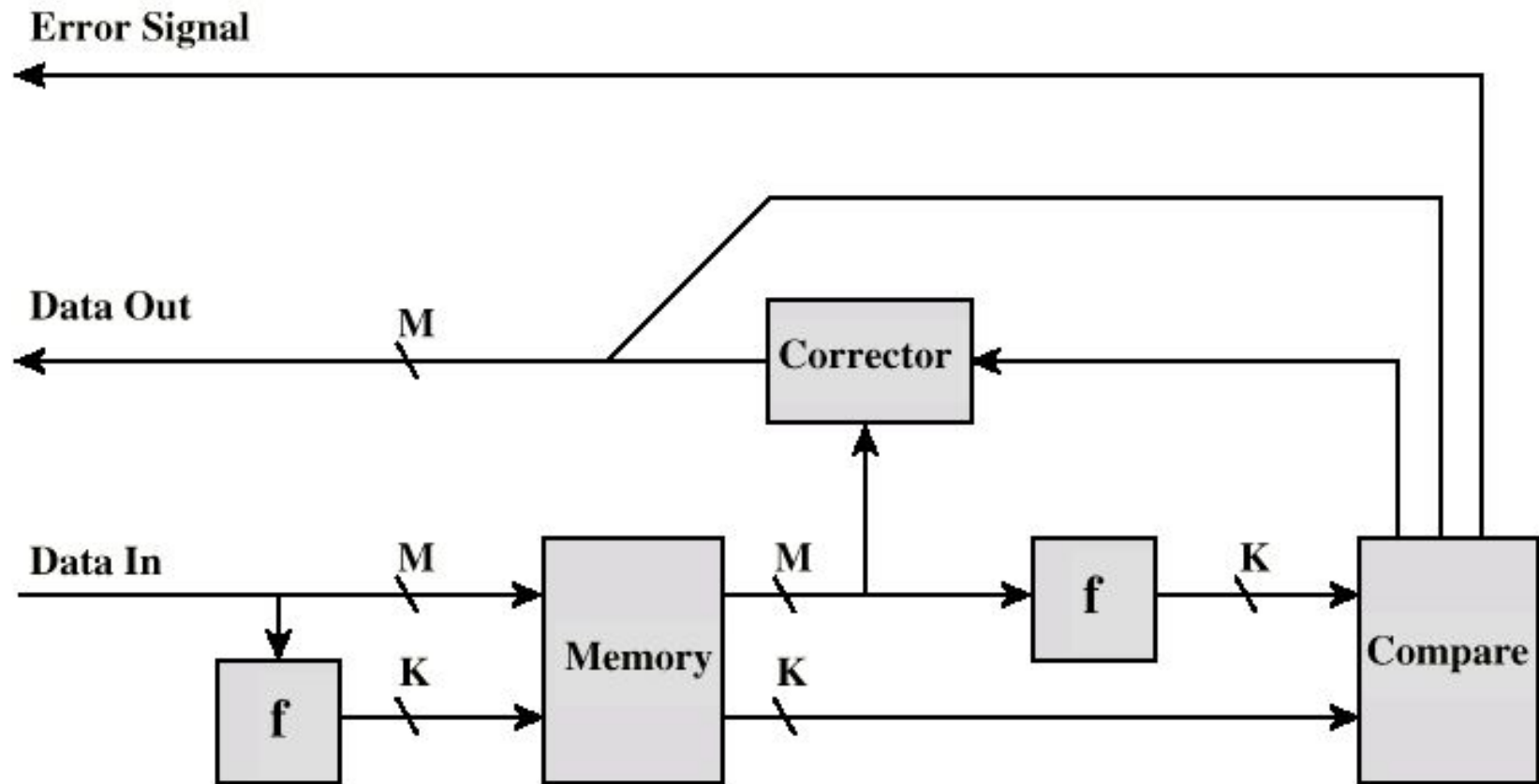
Error Correction

- Hard Failure
 - Permanent defect
- Soft Error
 - Random, non-destructive
 - No permanent damage to memory
- Detected using Hamming error correcting code

Cont.

- No errors are detected. The fetched data bits are sent out.
- An error is detected, and it is possible to correct the error. The data bits plus error correction bits are fed into a corrector, which produces a corrected set of M bits to be sent out.
- An error is detected, but it is not possible to correct it. This condition is reported.

Error Correcting Code Function



Thanks